Considering the Use of Stock and Flow Outcomes in Empirical Analyses: An Examination of Marriage Data

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Abstract

This paper fills an important void assessing how the use of stock outcomes as compared to flow outcomes may yield disparate results in empirical analyses, despite often being used interchangeably. We compare analyses using a stock outcome, marital status, to those using a flow outcome, entry into marriage, from the same dataset, the American Community Survey. This paper considers two different questions and econometric approaches using these alternative measures: the effect of the Affordable Care Act young adult provision on marriage using a difference-in-differences approach and the relationship between aggregate unemployment rates and marriage rates using a simpler ordinary least squares regression approach. Results from both analyses show stock and flow data yield divergent results in terms of sign and significance. Additional analyses suggest prior-period temporary shocks and migration may contribute to this discrepancy. These results suggest using caution when conducting analyses using stock data as they may produce false negative results or spurious false positive results, which could in turn give rise to misleading policy implications.

Keyword*:* Marriage, Data Quality, Unemployment, Business Cycle, Health Insurance, Affordable Care Act

JEL Classification: C18, J00, J12

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1 Introduction

Much empirical work in economics, as well as other disciplines, aims to evaluate how changes in policies and general conditions affect a wide range of individuals' choices, from employment and program participation to health insurance and marriage. This is a worthy endeavor, as understanding how a given factor affects individuals' behavior can be illustrative for understanding the full implications of changing policies and conditions as well as for future policymaking. However, research considering the same policy or condition often finds conflicting results. Since such analyses may use different data, sample criteria, analysis periods, and methods, it is difficult to identify the source of discrepancies.

This analysis considers one source of disparities in results of empirical analyses: the type of outcome measure used in the analysis – stock or flow. In most analyses assessing the effect of a policy change, we are most interested in seeing how behavior is affected by the policy, most often represented by a flow measure indicating entry or exit into the outcome of interest. However, stock measures are often used in conjunction with methods that examine changes in stock outcomes to proxy flow outcomes: considering the change in the proportion currently having the outcome of interest proxies net entry (entry less exit) while comparing the change in the proportion ever having the outcome of interest proxies entry.

Analyses using flow outcome measures tout the benefits of using these measures over stock outcome measures arguing that behavior and changes in policies and the labor market are more closely linked in flow data than in stock data (Abramowitz 2016; Abramowitz 2017; Bitler et al. 2004; Lichter et al. 2002; Schaller 2013) and that specification bias is minimized by using flow data (Klerman and Haider 2004). In addition, Buckles et al. (2011) found imprecise estimates associated with stock data as to compared to those from flow data, and accordingly, prefer their estimates from flow data. However, in many cases, only data representing the stock of the outcome of interest are available, and these stock outcomes are often used interchangeably with flow outcomes in empirical analyses. Further, work has shown that the main source of on marriage flows, administrative records, may be inferior to survey data under particular circumstances, for example, when a policy change of interest induces migration or misreporting (Blank et al. 2009). Within the literature on marriage alone, many papers use stock outcomes across pooled cross-sections. These include annual data, such as the Current Population Survey Annual Social and Economic Supplement (e.g., Abramowitz 2014; Lichter et al. 2002), as well as decennial data, such as samples from the U.S. Census (e.g., Goldin and Katz 2002; Gould and Paserman 2003; Loughran 2002; Wood 1995).

The effect of using stock outcome measures in lieu of flow outcome measures in empirical analyses has not been thoroughly considered. Klerman and Haider (2004) examined the effect of using flow outcomes as compared to stock outcomes to examine welfare caseload movements in the 1990s using California administrative data. They found that estimates using flow outcomes are more robust and larger than those obtained when using stock outcomes from the same data. Their findings are suggestive that the use of stock outcomes and related alternative empirical

specifications in prior literature estimating the relative importance of the economic expansion and welfare reforms in explaining the welfare caseload decline may explain how these studies reached widely varying conclusions regarding the cause of the decline. These findings suggest the value of considering the extent to which such measures and accompanying specifications may affect results across other subject areas as well as specific causes for concern and ways to mitigate them in the absence of flow data. In this analysis, we consider reasons for disparities in results across the two outcome measures and consider the implications.

To assess the use of stock and flow data in empirical analyses, this paper's analyses utilize questions on marital status, a stock measure, and marital history, a flow measure, in the American Community Survey (ACS). While the ACS is a cross-sectional survey, the survey's detailed questions on marital timing provide insight into changes in an individual's marital status over time and marriage flows in addition to stock measures of marital status. Accordingly, the ACS marriage data provide an opportunity to use a large nationally representative high-quality dataset to compare stock and flow outcome measures. While this analysis considers only marriage stocks and flows, the results are relevant for considering other outcomes that change over time, for example, educational attainment, debt, and others.

To consider how stock and flow data might yield different results, the paper explores two different analyses. First, we compare individual-level difference-in-differences-type analyses using marriage flow outcomes to those using marriage stock outcomes. To do this, we examine the effect of the 2010 Patient Protection and Affordable Care Act (ACA) young adult provision on marriage, re-estimating analysis performed by Abramowitz (2016). Next, we compare reported flow outcomes to those using constructed flow outcomes from stock outcomes. To do this, we consider the relationship between aggregate unemployment rates and marriage rates, re-estimating analysis performed by Schaller (2013) exploring this question using national vital statistics data and an aggregate-level ordinary least squares regression analysis.

Considering the effect of the ACA young adult provision on marriage, we first show that while estimates using marriage flow outcomes appear to be quite robust, estimates using marriage stock outcomes are not and can yield opposite-signed significant results with small changes to the specification of the estimating model. Next, considering the relationship between unemployment rates and marriage rates, we obtain disparate findings using marriage stock and marriage flow outcomes. For women ages 28–34, using marriage flow outcomes, we find a negative relationship between unemployment rates and marriage rates. When we use marriage stock outcomes for the same sample, we generally find much noisier estimates, suggesting that the issues with using marital status may limit statistical precision: we find results that are smaller in magnitude by up to 30% and become insignificant. Results of additional analyses support the hypotheses that prior-period temporary shocks and migration may contribute to this discrepancy. These results suggest using caution when conducting analyses using stock outcomes as they may produce false negative results or spurious false positive results, which could in turn give rise to misleading policy implications.

This paper makes several contributions. The paper assesses the use of stock and flow data in empirical analyses and identifies potential reasons for disparities between the two measures. While these analyses focus on marriage, the results generalize to other areas of interest. Results show the value of using flow data in analyses and, in turn, of collecting such data in surveys.

2 Conceptual Framework

The difference in two stocks over time should represent a flow. Since we often do not know how many individuals had a change in an outcome of interest - and instead only know the stock of individuals currently experiencing or having ever experienced the outcome - stock measures are often used with appropriate methods to proxy a flow. One way of approximating the effect on the flow is to use the stock outcome directly in an analysis. For example, using the stock measure as the outcome in a difference-in-differences analysis essentially compares changes in outcomes over time across treatment and control groups. Alternatively, one can approximate the effect on the flow by using the difference in two stocks to calculate a constructed flow measure.

This paper explores these two methods of using stock measures to approximate flow measures in empirical analyses and compares each stock measure to its corresponding flow measure. In the first method, which we refer to as the difference-in-differences approach, we compare over time the stock of individuals with a given outcome for different groups of interest, essentially identifying rates of entry into an outcome as the change in the proportion experiencing that outcome. In the second method, which we refer to as the constructed flows approach, we compare stocks over time for a particular cohort to construct a flow measure.

2.1 Difference-in-Differences Approach

Difference-in-differences approaches are generally used to examine the effect of a policy change on behavior, represented by entry or exit into an outcome (flow), but analyses often examine the change in stocks as proxies for this entry or exit. Considering the change in the proportion currently having the outcome of interest proxies net entry (entry less exit), which we will outline below for expositional purposes only, while comparing the change in the proportion ever having the outcome of interest proxies entry, which we will examine empirically in subsequent analyses.[1] If the necessary conditions for a valid difference-in-differences estimator are met, specifically, that the control and treatment groups exhibit parallel trends in the outcome of interest during the pre-treatment period, then, all else equal, the two approaches should yield qualitatively similar results. Estimating measures of entry, we can compare the two estimators as follows:

The DD estimator using a stock measure of the dependent variable takes the form:

$$\boldsymbol{DD_{stock} = \Delta Stock_{treatment} - \Delta Stock_{control}} \qquad (1)$$

which can also be written as

$$DD_{stock} = [Stock_{post} - Stock_{pre}]_{treatment} - [Stock_{post} - Stock_{pre}]_{control} \qquad (2)$$

[1] It is important to note that one limitation to examining entry as the change in two stocks representing ever having experienced the outcome of interest is that it limits analysis to outcomes involving a unique entry event or, if there are multiple entry events, only the first instance. Such an approach would not be suitable for examining multiple instances of entry, as in the case of remarriage or higher ordered births. However, such an approach is reasonable for examining unique or first entry events, as in the case of first marriage or first birth.

2.1.1 Using a Stock Measure Representing Currently Experiencing the Outcome of Interest

Given that using a stock measure that represents currently experiencing the outcome of interest represents:

$$CurrentStock_{post} = Entry_{post} - Exit_{post} + CurrentStock_{pre} \quad (3)$$

we can plug in and get:

$$DD_{CurrentStock} = [Entry_{post} - Exit_{post} + CurrentStock_{pre} - CurrentStock_{pre}]_{treatment} - [Entry_{post} - Exit_{post} + CurrentStock_{pre} - CurrentStock_{pre}]_{control} \quad (4)$$

For a measure of entry, assuming that the difference in the stocks over the pre- and post-treatment periods can only be attributed to entry in the post-treatment period and that the exit rates across the treatment and control groups are the same, we get:

$$DD_{CurrentStock} = Entry_{post,treatment} - Entry_{post,control} \quad (5)$$

The same logic holds for constructing a measure of net entry, except we do not need to assume the exit rates between the treatment and control groups are the same.

2.1.2 Using a Stock Measure Representing Ever Having Experienced the Outcome of Interest

Alternatively, we can examine entry as the change in two stocks representing ever having experienced the outcome of interest:

$$Entry_{post} = EverStock_{post} - EverStock_{pre} \quad (6)$$

And rearranging gives:

$$EverStock_{post} = Entry_{post} + EverStock_{pre} \quad (7)$$

And we can then plug in and get

$$DD_{EverStock} = [Entry_{post} + EverStock_{pre} - EverStock_{pre}]_{treatment} - [Entry_{post} + EverStock_{pre} - EverStock_{pre}]_{control} \quad (8)$$

Then, for a measure of entry using a stock representing ever having experienced the outcome of interest, we again do not have to assume the exit rates across the treatment and control groups are the same, and assume only that the difference in the stocks over the pre- and post-treatment periods can only be attributed to entry in the post-treatment period, we get:

$$DD_{EverStock} = Entry_{post,treatment} - Entry_{post,control} \quad (9)$$

We can then compare the DD estimator using a stock measure representing ever having the outcome of interest to the DD estimator using a flow measure representing entry into the outcome of interest. The DD estimator using a flow measure of entry into the outcome of interest takes the form:

$$DD_{flow} = \Delta Entry_{treatment} - \Delta Entry_{control} \qquad (10)$$

Rewriting gives:

$$DD_{flow} = [Entry_{post} - Entry_{pre}]_{treatment} - [Entry_{post} - Entry_{pre}]_{control} \qquad (11)$$

Rearranging gives:

$$DD_{flow} = [Entry_{post,treatment} - Entry_{post,control}] - [Entry_{pre,treatment} - Entry_{pre,control}] \qquad (12)$$

Assuming the control and treatment groups exhibit parallel trends in the outcome of interest during the pre-treatment period implies that the flow and stock difference-in-difference estimators are equivalent:

$$DD_{flow} = Entry_{post,treatment} - Entry_{post,control} = DD_{stock} \qquad (13)$$

It follows that, assuming the parallel trends assumption is met, the alternative outcome measures can be used interchangeably.

While this decomposition shows how the stock and entry measures may yield equivalent results in a difference-in-differences framework, there are several reasons why they may not. One reason is that discrepancies in results of alternate measures may reflect model misspecification rather than the outcome measures themselves and differences-in-differences approaches tend to be particularly prone to model misspecification. Another reason, particular to difference-in-differences specifications, is that the difference-in-differences approach assumes entry does not vary differentially by cohort, but this is often not a testable assumption. The problem arises because difference-in-differences analyses examine changes in outcomes over time, and the composition of the analysis sample necessarily changes over time as the same individuals are not followed over time, but rather outcomes for successive samples are compared over time. This issue is most pronounced in the case of differential prior period shocks for treatment and control groups: for example, if a prior period shock affected outcomes for a specific cohort of individuals differentially across the treatment and control groups in a period before they appear in the analysis sample, we may attribute effects of the prior period shock to the policy change of interest. In this case, the relationship in Equation 6 does not hold. Instead, we would have:

$$EverStock_{post} = Entry_{post} + EverStock_{pre} + PriorShock \qquad (14)$$

If the shocks across treatment and control groups are the same, they will cancel out in the difference-in-differences estimation. Otherwise, the net effect of the shock on the treatment as compared to the control group will be attributed to the policy change of interest. Since the flow measure captures entry but not prior shocks, results of analyses using these two measures would diverge.

2.2 Constructed Flow Approach

To consider the use of alternate measures in a simpler framework less prone to model misspecification and free from prior-period temporary shocks, we construct flow outcomes from stock measures by comparing outcomes for successive ages over time and then compare these constructed flow outcomes to reported flow outcomes using an ordinary least squares framework.

For this constructed flow approach, the reported rate takes the current year *y* proportion of individuals reporting entry into an outcome for each age *a:*

$$Entry_Reported_{ya} = \frac{OutcomeEntry_{ya}}{Pop_{ya}} \qquad (15)$$

For the constructed measure of entry, we calculate entry as the difference in the proportion *ever having the outcome* between the prior year and the current year. Specifically, we estimate the change in the proportion ever having the outcome by subtracting the prior year *y-1* proportion ever having the outcome for a given age *a-1* from the current year *y* proportion for the subsequent age *a*:

$$Entry_Constructed_{ya} = \frac{EverOutcome_{ya}}{Pop_{ya}} - \frac{EverOutcome_{y-1,a-1}}{Pop_{y-1,a-1}} \qquad (16)$$

The use of these alternate entry measures in an ordinary least squares framework simulates the differences-in-differences approach comparison with less possibility for model misspecification since the interpretation of the control variables should be the same using either outcome and the use of contemporaneous control variables should not be problematic. The estimates should also not reflect cohort-specific shocks since the same cohorts are compared over time.

3 Methods

To consider how stock and flow data might yield different results, the paper explores two different analyses examining marriage. Our analyses will compare the use of flows (reported marriage rates) and stocks (marital status) as outcome measures in the difference-in-differences approach and the use of flows (reported marriage rates) and flows constructed from stocks (constructed marriage rates) in the cohort approach. First, we compare an individual-level difference-in-differences-type analysis using ACS marital history data to ACS marital status data. To do this, we examine the effect of the 2010 Patient Protection and Affordable Care Act (ACA) young adult provision on marriage, re-estimating analysis performed by Abramowitz (2016). Next, we consider an aggregate-level ordinary least squares regression analysis where we compare ACS reported marital history data and ACS constructed marriage rates from stock data. To do this, we consider the relationship between aggregate unemployment rates and marriage rates, re-estimating analysis performed by Schaller (2013) exploring this question using national vital statistics data.

4 Data

This paper's analyses utilize new questions on marital history in the ACS to assess the use of stock and flow data for empirical analyses. The ACS began asking more detailed questions about

individuals' marriage histories beginning in 2008 (Department of Health and Human Services 2008) in addition to marital status, which was collected in prior years to replace NCHS's discontinued collection of detailed marriage statistics as part of the vital statistics in 1996 due to concerns about data quality (O'Connell et al. 2007). The data collected include year of most recent marriage, number of times married, and whether the respondent married, divorced, or was widowed in the prior 12 months. Thus, while the ACS is a cross-sectional survey, the survey's detailed questions on marital timing provide insight into changes in an individual's marital status over time and marriage flows in addition to stock measures of marital status. The ACS estimates of rates of marriage, divorce, and widowhood have been found to be comparable to vital records tabulations at the state and national levels (Elliott, Simmons, and Lewis 2010).[2] Accordingly, the ACS provides an opportunity to use a large nationally representative high-quality dataset to compare cohort reports of marital status to marital history information.

To compare analyses between the ACS marital history data and the ACS marital status data, we use marital history data from the 2008–2013 internal ACS one-year data and marital status data from the 2007–2013 internal ACS one-year data.[3] The ACS is a nationwide survey conducted continuously throughout each year. It is conducted in all U.S. counties and Puerto Rico municipos. Over 3 million housing unit addresses are sampled annually, in addition to a sample of individuals living in group quarters, such as college dormitories. The internal ACS data provide a larger sample than public use data (U.S. Census Bureau, 2015).

This paper compares the use of three alternative sources of information on marriage outcomes in the ACS, two from marital history data and one from data on marital status.[4] The marital history outcomes include whether respondents married in the calendar year prior to the survey[5] and whether respondents married, divorced, or were widowed in the 12 months prior to their survey response. For all marital history variables, further information on the number of times married permits identification of first marriages separately from subsequent marriages. The marital status variable includes information on the respondent's current marital status, which we use to calculate

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[2] Despite the ACS and vital statistics producing similar divorce rates, Kennedy and Ruggles (2014) argue that the ACS may be a more accurate measure of divorce than the vital statistics.

[3] The ACS began asking about marital history in 2008. Since the questions are used to estimate marriage rates in the prior calendar year or the 12 months prior to the survey response, marital history outcomes cover marriage rates over 2007 through 2012 in this analysis. Marital status outcomes are calculated for a given year as the difference in the relevant proportion married for that year from the subsequent year. Accordingly, we use marital status information from 2007 data to calculate marriage rates over the 2007 calendar year (the difference between the 2008 and 2007 proportions), comparable to the 2008 marital history data capturing whether a respondent married in the prior calendar year.

[4] Prior to 2013, individuals reporting being married to a same-sex partner were classified as "unmarried partners." Beginning in 2013, ACS data and data products include same-sex married couples along with all married couples. This analysis uses the marital status classifications as provided in the data. Analyses retaining the marital status of individuals reporting being married to a same-sex partner for 2012 and earlier yield quantitatively similar results.

[5] Information on the year of most recent marriage is used to identify whether the respondent married in the calendar year prior to the survey. Since the ACS collects data over the course of the calendar year, the outcome measure is defined as the prior rather than the current calendar year in order to capture all marriages occurring throughout the year. Given the short timeframe over which we examine marriage behavior (the prior calendar year), measurement error associated with the lookback period should be minimal.

the proportion of the population ever married. Comparing this calculation over time simulates a marriage rate.

We would expect to see larger measurement error associated with the simulated marriage rate measure both due to its construction as a difference of two estimates as well as variations in the samples for each estimate. However, this feature of the simulated marriage rate measure should bias estimates toward zero, making analyses less likely to produce significant results.

5 The Effect of the Affordable Care Act Young Adult Provision on Marriage Rates – Difference-in-Differences Approach

In difference-in-differences approaches, stock measures are commonly used as outcome measures due to the limited availability of flow outcome data. In this section, we consider the effect of using these alternate outcome measures in such a framework.

For this approach, we re-estimate the analysis performed by Abramowitz (2016) examining the effect of the ACA young adult provision on marriage rates. As of 2010, this provision allowed 19-25 year-olds to be covered by their parents' private health insurance plans regardless of marital status, student status, and whether they have children. The analysis explores whether this new avenue for obtaining coverage outside of marriage led to a decrease in marriage by applying difference-in-differences-type methods comparing outcomes for individuals in the treatment age group (ages 23-25) to outcomes for similarly aged individuals at ages not covered by the provision (ages 28-30) before and after provision enactment and implementation.[6] We estimate analyses using marital history and marital status outcomes and consider alternative control variables to examine the robustness of the results with each type of outcome.

Since this framework identifies changes across a treatment and a control group generally over time, before and after a policy, it is possible to use individual-level marital status as an outcome variable and consider the coefficient estimate on the outcome of interest as the change in the proportion married attributable to the policy, like a rate of entry into marriage.

5.1 Approach

The analysis sample includes individuals aged 23-25 and 28-30 at the time of survey response from all U.S. states and the District of Columbia living in households or college dormitories. We include only individuals born in the United States to exclude any effects of in-migration, though results included foreign-born individuals show similar patterns. Active duty military are also excluded from the sample. For the flow analysis, since the outcome of interest is whether an individual married in the calendar year prior to the survey year, only individuals who were unmarried in the calendar year prior to the survey year and those who married in the calendar year prior to the survey year are included in these analyses.[7] For the stock analysis, all individuals are included.

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[6] Individuals aged 26 and 27 were excluded to identify clear effects for the treatment age group of 23-25.
[7] Analyses including all individuals yield quantitatively similar results.

Data from the 2007 or 2008 through 2013 survey years are used to examine the periods before, during, and after provision enactment and implementation.[8] The enactment period covers the time after the provision was enacted, but before insurance companies were required to implement it, during which some companies voluntarily extended coverage to young adults; the implementation period covers the time insurance companies were required to implement the provision. Since the outcome of interest is marriage in the calendar year prior to the survey year, provision enactment and implementation periods are extrapolated to cover corresponding calendar years. For the purposes of the flow analysis, the period before provision enactment covers the 2008 through 2010 survey years (2007 through 2009 calendar years), the enactment period covers the 2011 survey year (2010 calendar year); and the implementation period covers the 2012 through 2013 survey years (2011 and 2012 calendar years). For the purposes of the stock analysis, the period before provision enactment covers the 2007 through 2009 survey years (2007 through 2009 calendar years), the enactment period covers the 2010 survey year (2010 calendar year); and the implementation period covers the 2011 through 2013 survey years (2011 through 2013 calendar years).

Comparing the proportion ever married, in Fig. 1, and reported first marriage rates and changes in the proportion married, Fig. 2, for ages 23-25 (treatment) and 28-30 (control) shows concerning disparities. In Fig. 1, we see similar trends in the proportion ever married for both age groups, with the proportion declining each year. However, in Fig. 2, we see quite the opposite: while first marriage rates of 23-25-year-olds do decline each year, those for 28-30-year-olds remain stable over the period, even increasing in some years. This represents the basis for the DD estimator using a flow measure of entry as in Eq. 10. Comparing the change in the proportion ever married, as in a difference-in-differences approach, we actually see a steady increase in the change for 23-25-year-olds, while the results for 28-30-year-olds are much less stable. This represents the basis for the DD estimator using a stock measure of entry as in Eq. 9 using the relationship in Eq. 6. Given that we see such divergent trends in the data, we would expect to see divergent results in any analyses using the alternate measures. We next examine whether this is indeed the case after controlling for other characteristics that may affect the outcome variables.

The following difference-in-differences-type regression model estimates the effects of the provision on marriage:

$$y_{iajt} = \beta_0 + \beta_1 Enact_{ajt} + \beta_2 Implement_{ajt} + \beta_3\left(TreatedAge_a \times Enact_{ajt}\right) + \beta_4\left(TreatedAge_a \times Implement_{ajt}\right) + \beta_5 State_j + \beta_6 Year_t + \beta_7 TreatedAge_a + \beta_8 X_i + \beta_9 Z_{jt} + \varepsilon_{iajt} \quad (17)$$

where y_{iajt} is a dummy variable equal to one if an individual *i* of age group *a* living in state *j* during survey year *t+1* reports having first married in the prior calendar year *t* and is equal to zero otherwise for the marital history analysis and is a dummy variable equal to one if an individual *i* of age group *a* living in state *j* during survey year *t* reports having ever married and is equal to zero otherwise for the marital status analysis. The coefficients on the interaction terms β_3 and β_4 capture

[8] We need to include an additional year of data for the analysis using stock outcomes to represent an equivalent time period as in the analysis using flow outcomes. Results are quantitatively similar if the 2007 data is dropped from the analysis using stock outcomes.

the associated difference in the probability of each outcome for a person in the treatment age group during the provision enactment and implementation periods, respectively, relative to a person in the control age group, or a person in the treatment age group in the pre-enactment period, holding other characteristics constant. Vectors of parameters are included to control for state fixed effects (β_5), year fixed effects (β_6), and treatment age-group fixed effects (β_7). The X vector (β_8) controls for individual demographic characteristics including dummy variables for age, sex, race/ethnicity, and student status. The Z vector (β_9) controls for state-year-level conditions including the age-specific state-year sex ratio and unemployment rate, the interaction of the unemployment rate with the control for the treatment age group, as well as a national linear calendar year trend and state-calendar year trends. The error term is represented by ε. In all tables and regressions, the data are weighted to be population-representative using population weights. As per Bertrand *et al.* (2004), in all regressions, White robust standard errors clustered by state are used to control for serial correlation among the outcomes and the policy changes of interest. To test the sensitivity of the results, we also estimate specifications omitting the interaction of the unemployment rate with the control for the treatment age group.

5.2 Regression Results

Table 1 shows the results of the difference-in-differences analyses. Table 1 first replicates the analysis of Abramowitz (2016) using a flow measure of marriage, whether an individual first married in the prior calendar year, and finds 0.51 percentage point and 0.52 percentage point decreases in first marriage rates associated with provision enactment and implementation, respectively.[9] Performing the analysis using a stock measure of marriage, whether an individual has ever been married, yields results that are much smaller in magnitude and not significant. For the full sample, we see 0.02 percentage point and 0.32 percentage point decreases in the proportion married associated with provision enactment and implementation, respectively. To examine the robustness of the results of each analysis, we make small changes to the specification, first omitting the interaction of the unemployment rate with the control for the treatment age group and then omitting the unemployment rate control entirely, as well as its interaction term, presented subsequently in Table 1. For the flow analysis, we see results that are slightly smaller in magnitude but similar significance for both specifications. However, for the stock analysis, estimates for the specification with no interaction term are opposite signed but insignificant, and most dramatically, estimates for the specification with no unemployment rate control or interaction term are opposite-signed and significant.

The divergent regression results are not surprising given the disparate patterns seen in the data in Fig. 1 and Fig. 2. One potential explanation for such a divergence could be related to past temporary shocks affecting only the treatment or control group before they reached the ages included in the sample. Evidence of these shocks would not appear in analyses considering parallel trends since they affect individuals before they are included in the sample. They should also not affect flows directly, since they are past temporary shocks. However, they could affect the stock

[9] Abramowitz (2016) looks at the marriage rate; we look at the first marriage rate here to be consistent with the marital status measure since a change in the proportion ever married should simulate a first marriage rate. Results are generally consistent, which is not surprising since the vast majority of marriages for people at these ages are first marriages.

of individuals ever experiencing the outcome of interest. This could, in turn, explain disparate results of analyses using stock outcomes as compared to those using flow outcomes.

6 The Relationship between Unemployment Rates and Marriage Rates – Constructed Flows Approach

Since the above results might be due to the specifics of the difference-in-differences specifications, we next explore a simpler ordinary least squares specification. Since it is not comparable to use marriage stocks as outcomes in the ordinary least squares regressions directly, here we compare outcomes for reported and constructed marriage rates.

6.1 Approach

We consider how stock and flow data might yield different results in a simpler ordinary least squares framework by exploring the relationship between state-level unemployment rates and state-level marriage rates. Schaller (2013) finds that increased unemployment rates are associated with a reduction in entry into marriage using aggregate 1978-2009 national vital statistics data. This analysis re-estimates the analysis performed by Schaller (2013) for women ages 28–34 using 2008-2013 ACS marital history data and 2007-2013 ACS marital status data.[10] We examine outcomes for women ages 28–34 because we see the most pronounced between unemployment rates and marriage rates for this age group.[11] We restrict the sample to the civilian household population and the population living in college dormitories from all U.S. states and the District of Columbia. We include only women to avoid double-counting marriages and because women are more likely to have reported rather than imputed responses relative to men (O'Connell 2006; O'Connell et al. 2007).

Comparing reported and constructed first marriage rates over 2011 to 2012 by state in Table 2 shows that while some estimates are quite similar, many estimates tend to be drastically different in magnitude with a number statistically significantly different.

We first compare constructed and reported marriage rates for women ages 28-34 at the national level as well as the U.S. annual unemployment rate in Fig. 3 for 2008-2013. We include only individuals born in the United States to exclude any effects of in-migration, though results included foreign-born individuals show similar patterns. In general, the ACS reported rates show similar trends. As the unemployment rate rose during the recessionary period over 2007–2010, the marriage rate fell. Concurrent with a fall in the unemployment rate over 2010–2012, marriage rates rose for 28–34-year-olds over 2011–2012. While the constructed rates show the same overall trends as the reported rates, they exhibit considerable variation in individual years.

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[10] As would be expected since the ACS and vital statistics both capture information on U.S. marriages, results from analyses comparing ACS marital history and vital statistics data over 2007-2012 including women of all ages using vital statistics data from 2007 to 2012, presented in Appendix 1, yield qualitatively similar results.

[11] Results of analyses by age group are presented in Appendix 2.

Having seen variation in constructed and reported marriage rates at the state and national levels, we consider these measures more thoroughly in a regression framework. We estimate the following baseline specification:

$$\boldsymbol{M_{jt} = \beta_0 + \beta_1 U_{jt} + \beta_2 State_j + \beta_3 Year_t + \beta_4 State_j \text{ X } Year_t + \beta_5 X_{jt} + \varepsilon_{jt}} \qquad (18)$$

where M_{jt} represents the natural log of the marriage rate for state j during year t and U_{jt} represents the unemployment rate for state j during year t.[12] Using marital history data, these outcomes are calculated as the prior calendar year first marriage rate and prior 12 months first marriage rate out of the unmarried female population for each state-year. Using marital status data, this outcome is calculated as the difference in the proportion ever married for 28–34-year-olds in the current calendar year and 27–33-year-olds in the prior calendar year, divided by the proportion never married in the prior calendar for each state-year. We use the Bureau of Labor Statistics Local Area Unemployment Statistics state-year unemployment rate.[13] We include state fixed effects, year fixed effects, and state-specific linear time trends. The X vector controls for state-year demographic characteristics including the proportion Black, proportion Hispanic, proportion with a high school degree only, proportion with a college degree, calculated using the analysis sample. The error term is represented by ε. In all regressions, the data are weighted to be population-representative of the unmarried female population ages 28–34 for each state and year in the analysis sample.[14] We estimate White robust standard errors clustered by state.

6.2 Regression Results

Results considering the effect of unemployment rates on both log first marriage rates and first marriage rates are presented in Table 3. We estimate effects on both the log rate and the rate to consider the compare results including negative observations since the marriage rate calculated using marital status data can be negative, and these observations are dropped when taking logs.[15] Using marital history data, the prior calendar year and prior 12 months outcomes show similar magnitude and significance of decreases of 4.0% and 3.7%, respectively, in the first marriage rate associated with a one percentage point increase in the unemployment rate. Both log rates and rates show a significant negative effect of unemployment rates. For the constructed outcome calculated using marital status data, using the log rate, we see a result that is similar in magnitude and not significant: a one percentage point increase in the unemployment rate is associated with a 2.7% decrease in marriage rates. Using the actual rate shows a positive result that is not significant.

While these differential results for marriage stock and marriage flow data are concerning, they should not be problematic for empirical analyses if they only reflect measurement error and result in non-significant results.

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[12] For the ACS analyses, marriage rates are calculated using survey year *t+1* data for respondents reporting having married in the prior calendar year or married in the prior 12 months as a proportion of the unmarried female population.
[13] For analyses of current or prior calendar year measures, we use the unemployment rate associated with the calendar year. For analyses of prior 12 months and marital status measures, we use the average unemployment rate of the survey year and the prior calendar year.
[14] This is consistent with the methodology used by Schaller (2013).
[15] Taking the logs results in approximately 46 observations (about 15% of the sample) being dropped.

To consider one avenue that may affect the differential results for flow as compared to stock measures of marriage, we examine migration. Comparing year-over-year proportions married is problematic if single and married individuals move across states at different rates. For example, even if there is no change in the proportion of single people marrying in a given state, if more married people move in or out of that state, the constructed marriage rate for that state will change. Indeed, Abramowitz and Dillender (2016) find that, in general, across all ages, migration tends to jump sharply in the year of marriage and the following year, and then decreases gradually with time since marriage. They also find that net migration rates differ significantly by state and marital status, and in particular, that the factors that influence these migration rates, such as economic conditions, may vary by marital status. For our analysis of the relationship between state unemployment rates and marriage rates, it follows that if individuals move differentially by marital status because of economic conditions, our estimated effect of the unemployment rate on marriage using a constructed measure of marriage reflects both a direct effect as well as changes in the outcome measure due to migration.

Since Abramowitz and Dillender (2016) find that a large proportion of women move across states around their time of marriage, we try to isolate the effects of migration on our estimates. Accordingly, we re-estimate our regression results for the relationship between state-year unemployment and first marriage rates for women ages 28-34 using our marital history outcome measure, but first, we restrict our sample to only women living in the United States one year prior, and then we perform the analysis with the state defined as state of residence one year prior. Results, presented in Table 4, suggest effects that are larger in magnitude. This is not surprising, since conditions prior to marriage, rather than after marriage, are more likely to influence the marriage decision.

7 Discussion

This paper assessed the use of stock and flow data in empirical analyses. The paper considered both individual-level difference-in-differences analyses, as well as aggregate-level ordinary least squares regression analyses. Considering the effect of the ACA young adult provision on marriage in a difference-in-differences framework, we showed that while estimates using marriage flow outcomes appear to be quite robust, estimates using marriage stock outcomes are not and can yield opposite-signed significant results with small changes to the specification of the estimating model. Considering the relationship between unemployment rates and marriage rates in an ordinary least squares framework, we also obtained disparate findings using marriage stock and marriage flow outcomes, finding much noisier estimates associated with the stock outcomes.

Results of additional analyses support the hypothesis that prior-period temporary shocks affecting only the treatment or control group before they reached the ages included in the sample may contribute to these discrepancies. Evidence of these shocks would not appear in analyses considering parallel trends since they affect individuals before they are included in the sample. They should also not affect flows directly, since they are past temporary shocks. However, they could affect the stock of individuals ever experiencing the outcome of interest.

Results of other research also support the hypothesis that differential migration patterns correlated with the outcome of interest may contribute to these discrepancies. Abramowitz and Dillender

(2016) find that, in general, across all ages, migration tends to jump sharply in the year of marriage and the following year, and then decreases gradually with time since marriage. They also find that net migration rates differ significantly by state and marital status, and in particular, that the factors that influence these migration rates, such as economic conditions, may vary by marital status. For our analysis of the relationship between state unemployment rates and marriage rates, it follows that if individuals move differentially by marital status because of economic conditions, our estimated effect of the unemployment rate on marriage using a constructed measure of marriage reflects both a direct effect as well as changes in the outcome measure due to migration. The analyses performed in this paper to isolate the effects of migration on our estimates show the value in using geographic measures reflecting the period relevant to the analysis, and in turn to collecting these data in surveys or using administrative records, and suggest caution for using less aligned measures.

Our findings also underscore the point that using a cumulative or path-dependent stock outcome instead of a flow outcome requires alternative interpretations of the controls used in the specification. While the coefficients estimated for the other controls used in flow estimation represent the effect of the control on entry, in the stock estimation they represent the effect of the control on the stock, and these two effects could be quite different if the stock is path dependent. Further, in difference-in differences estimations using a stock outcome that is path dependent, the inclusion of contemporaneous controls is inappropriate, since current conditions cannot directly affect the likelihood of the outcome in prior periods.

The results of this analysis suggest a number of avenues for future work. The mechanisms by which prior-period temporary shocks may affect path-dependent stock outcomes warrants more consideration. Future work could also consider more thoroughly the characteristics of outcomes likely to be affected by the use of stock as compared to flow outcome measures and ways to evaluate or mitigate such effects when only data on stock outcomes is available.

8 Compliance with Ethical Standards

The authors declare that they have no conflict of interest.

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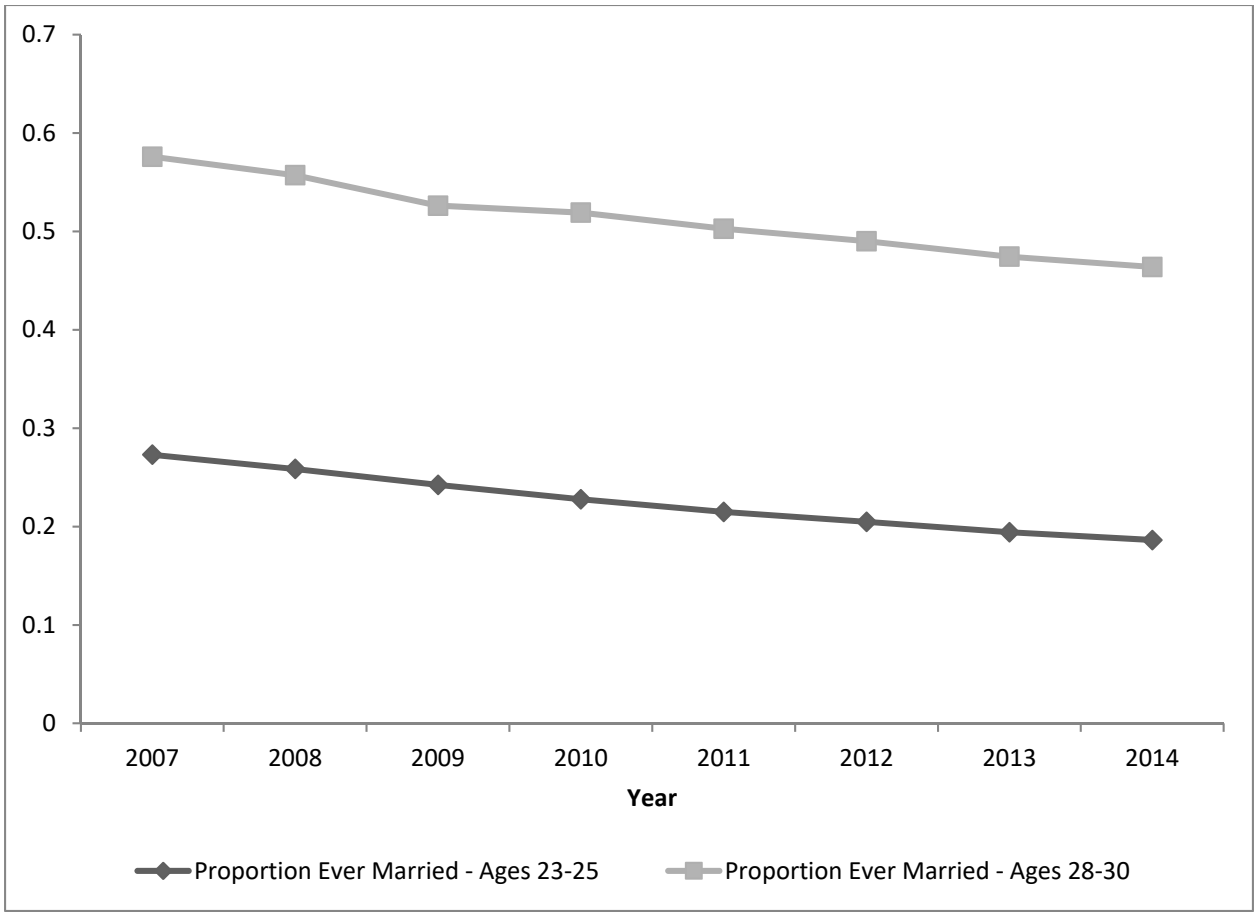
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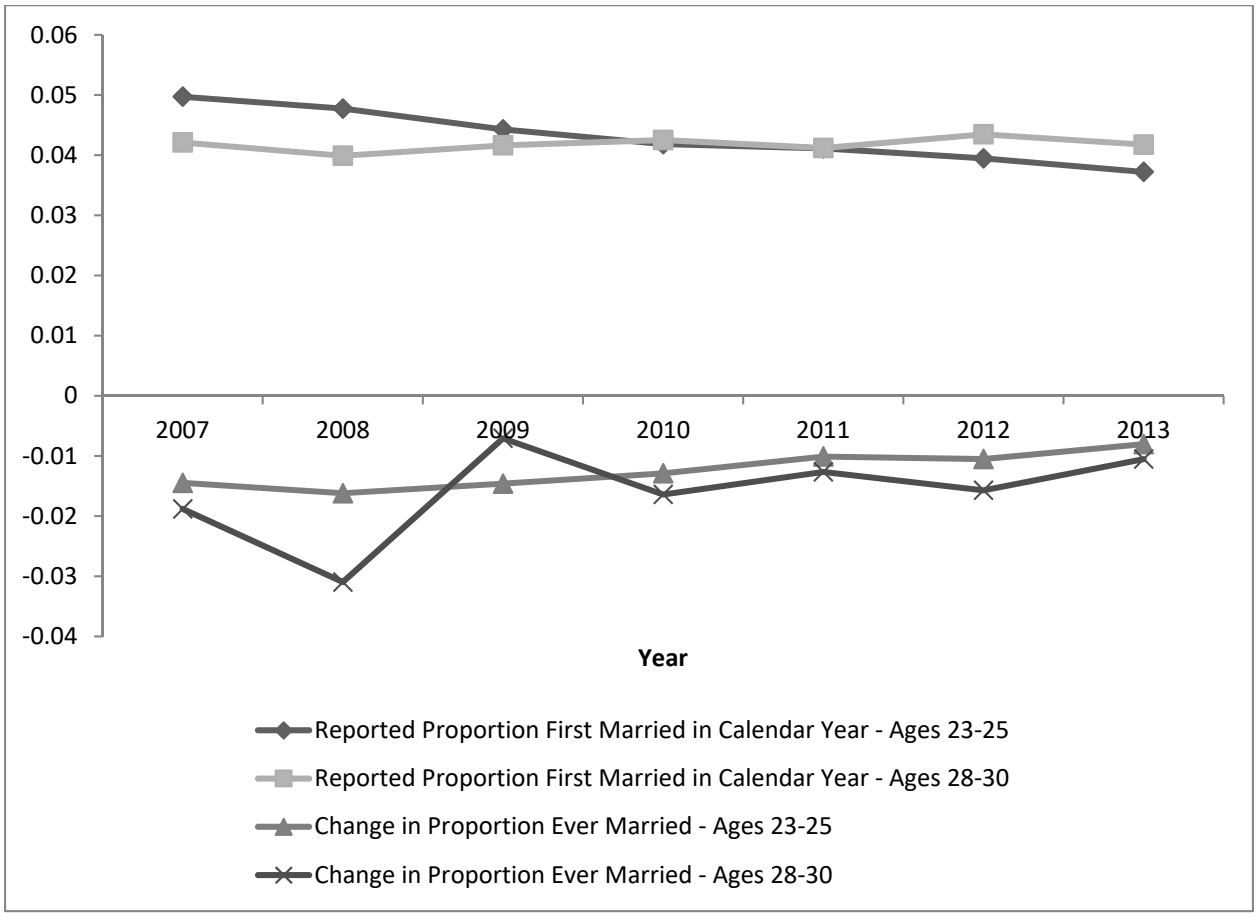
Figures

Fig. 1 Proportion Ever Married, 2007-2014, Ages 23-35 and 28-30



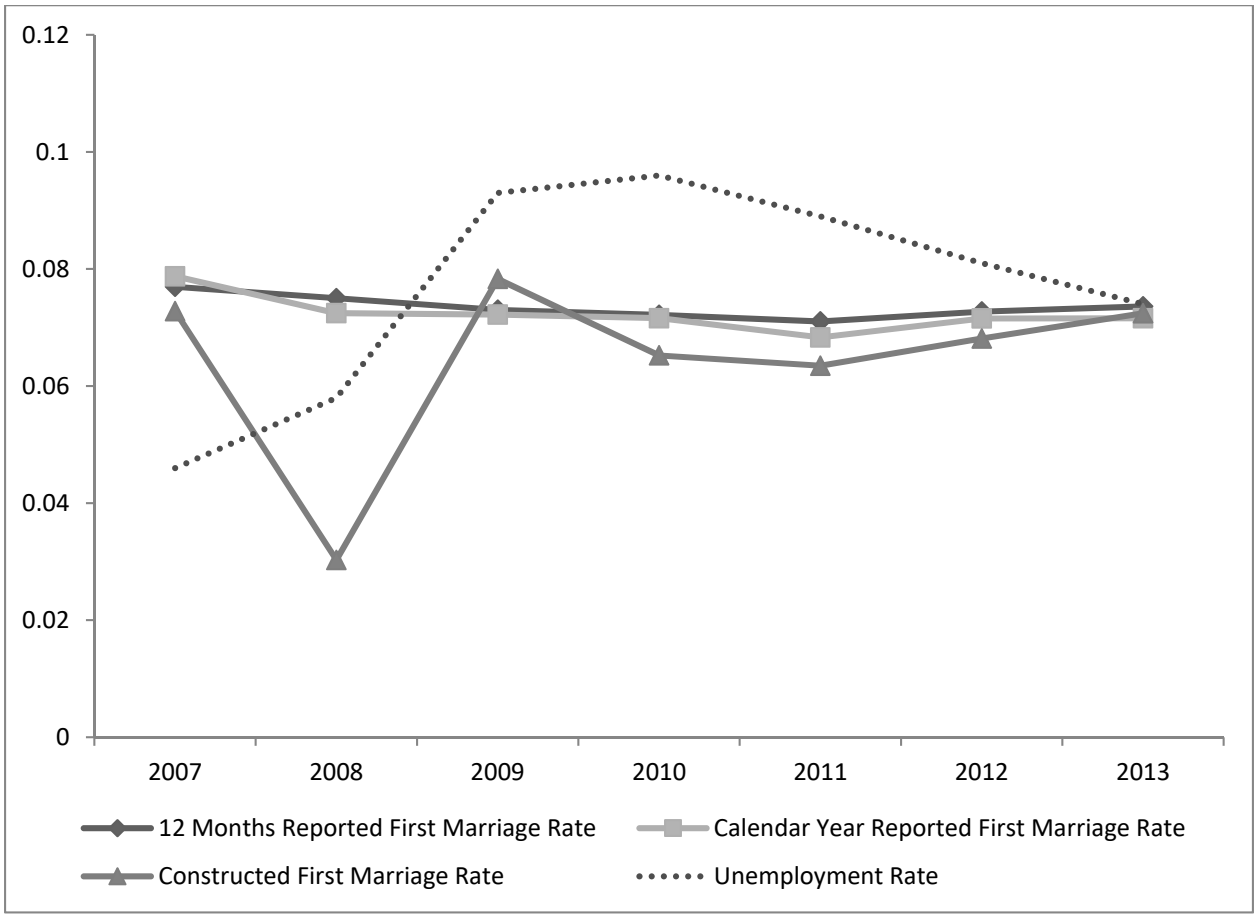
SOURCE: 2007-2014 1-year ACS data.

Fig. 2 Reported First Marriage Rates and Change in Proportion Ever Married, Ages 23-25 and 28-30



SOURCE: 2007-2014 1-year ACS data.

Fig. 3 State-Year Unemployment Rates and Constructed and Reported First Marriage Rates for Women Aged 28-34



SOURCE: 2007-2014 1-year ACS data.

Tables

Table 1 Regression Results for Relationship between ACA Young Adult Provision and Marriage

	Unemployment Interacted with Age Group		No Interaction		No Unemployment Control	
	First Married in Prior Calendar Year	Ever Married	First Married in Prior Calendar Year	Ever Married	First Married in Prior Calendar Year	Ever Married
Treatment*Enactment	-0.0051**	-0.0002	-0.0037**	0.0052	-0.0042***	0.0073**
	(0.0024)	(0.0063)	(0.0016)	(0.0032)	(0.0015)	(0.0031)
Treatment*Implementation	-0.0052***	-0.0032	-0.0042***	0.0023	-0.0050***	0.0060***
	(0.0018)	(0.0063)	(0.0013)	(0.0021)	(0.0011)	(0.0019)
Observations	1,247,472	1,903,879	1,247,472	1,903,879	1,247,472	1,903,879

SOURCE: 2008 through 2013 1-year ACS data.
NOTES: Shown are coefficients estimated from the specification of the determinants of the probability of having first married in the prior 12 months and the probability of having ever married. Each column presents results from a single regression. Regressions are estimated using population weights. Specifications include state, year, and age fixed effects, national and state-year trends, as well as individual demographic controls, the age-specific state-year sex ratio and unemployment rate, and an interaction of the treatment age group and the age-specific unemployment rate, except as noted. The first marriage sample includes all native-born individuals who were unmarried in the calendar year prior to the survey year and those who married in the calendar year prior to the survey year ages 23-25 and 28-30. The ever married analysis sample includes all native-born individuals ages 23-25 and 28-30. Robust standard errors clustered at the state level in parentheses, *** p<0.01, ** p<0.05, * p<0.1.

Table 2 2011-2012 Constructed and Reported Marriage Rates for 25-Year Old Women by State

State	N 2011	N 2012	Constructed First Marriage Rate (Percent)	S.E.	ACS Married First Time in Past Year (Percent)	S.E.
Alabama	419	453	6.4	4.0	6	1.4
Alaska	131	133	-4.6	8.1	5.4	3.4
Arizona	396	561	-5.2	4.2	4.3**	1.1
Arkansas	217	287	4.8	6.2	4.8	1.4
California	3,095	3,661	6.2	1.4	4.4	0.4
Colorado	444	504	12.0	3.6	6.1	1.4
Connecticut	294	276	3.9	4.3	6.1	1.5
Delaware	79	83	11.1	8.6	4.7	3.0
District of Columbia	109	126	4.1	3.1	3.6	2.0
Florida	943	1,309	6.1	2.4	4.4	0.6
Georgia	643	796	9.6	3.0	4.7	0.7
Hawaii	122	147	11.0	6.6	6.5	2.7
Idaho	130	132	13.6	7.5	10.2	3.6
Illinois	1,190	1,323	5.3	2.0	4.4	0.6
Indiana	602	609	5.4	3.6	6	1.1
Iowa	346	365	6.3	4.9	3.9	1.0
Kansas	319	314	15.3	4.7	5.7*	1.6
Kentucky	330	409	7.8	4.5	7	1.6
Louisiana	415	421	4.7	3.6	5.6	1.3
Maine	137	133	15.9	7.0	8.6	3.3
Maryland	478	526	1.7	2.8	3.4	0.9
Massachusetts	598	651	-1.1	2.5	3.4*	0.8
Michigan	1,036	1,072	4.9	2.1	7.3	1.0
Minnesota	706	750	7.7	3.2	9	1.6
Mississippi	222	228	11.1	5.6	1.9	0.7
Missouri	561	626	1.9	3.7	6.9	1.4
Montana	105	105	3.6	9.1	1.7	1.1
Nebraska	241	259	13.7	5.4	7.3	1.9
Nevada	157	244	6.5	6.2	3.3	1.0
New Hampshire	100	115	20.5	6.7	4.6**	2.2
New Jersey	689	711	1.2	2.6	5.1	0.8
New Mexico	165	202	5.5	6.4	7.7	2.2
New York	1,881	2,111	5.2	1.5	4.1	0.5
North Carolina	586	813	9.1	2.9	5.8	1.0
North Dakota	98	91	28.7	11.3	14.3	6.5
Ohio	1,020	1,133	5.7	2.3	5.9	0.7

Oklahoma	482	578	7.3	4.0	6.1	1.3
Oregon	289	311	-6.3	4.2	2.1*	1.0
Pennsylvania	1,359	1,444	7.0	2.2	5.2	0.6
Rhode Island	82	86	-1.8	6.7	4.8	2.5
South Carolina	295	409	2.0	4.0	3.9	0.8
South Dakota	90	114	16.3	10.1	10.5	5.2
Tennessee	433	533	-1.1	3.6	3.2	0.8
Texas	1,855	2,187	9.2	2.1	6.4	0.6
Utah	320	310	0.6	5.1	4.4	1.4
Vermont	86	96	22.6	7.4	2.3***	1.2
Virginia	638	742	11.5	3.3	5.2*	0.8
Washington	568	639	6.6	3.1	4.2	0.8
West Virginia	165	143	15.9	6.8	8.6	2.7
Wisconsin	701	758	1.2	3.1	7*	1.1
Wyoming	45	53	0.6	12.3	13.3	5.9

SOURCE: 2011 and 2012 1-year ACS data.
NOTES: *** p<0.01, ** p<0.05, * p<0.1 for significant difference from corresponding constructed estimate.

Table 3 Regression Results for Relationship between State-Year Unemployment and First Marriage Rates for Women Ages 28-34 Using the ACS

	Prior CY		Prior 12 Mo.		Constructed	
	Log	Rate	Log	Rate	Log	Rate
Unemployment rate	−0.040**	−3.303***	−0.037**	−3.497**	−0.027	1.485
	(0.015)	(1.222)	(0.018)	(1.405)	(0.107)	(6.569)
Observations	306	306	306	306	267	306
R-squared	0.691	0.688	0.682	0.658	0.433	0.311

SOURCE: 2007 through 2013 1-year ACS data.

NOTES: Shown are coefficients estimated from the specification of the determinants of state-year marriage rates. Each column presents results from a single regression. Unemployment rates come from the Bureau of Labor Statistics. Regressions are estimated using population weights. Specifications include state and year fixed effects, state-specific linear time trends, and state-year demographic controls. For the prior calendar year and prior 12 months estimates, the sample from which the marriage rate is estimated includes all unmarried women and women married in the prior calendar year or prior 12 months, as appropriate, ages 28–34 from the 2008 through 2013 1-year ACS data. For the constructed estimates, the sample from which the marriage rate is estimated includes all women ages 27–34 from the 2007 through 2013 1-year ACS data weighted by the proportion who were never married in the prior calendar year. Standard errors are clustered by state and shown in parentheses, *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 4 Regression Results for Relationship between State-Year Unemployment and First Marriage Rates for Women Ages 28-34

	State at Time of Survey, All	State at Time of Survey, Living in US 12 Months Ago	State 12 Months Ago
Unemployment Rate	-0.037**	-0.050***	-0.055***
	(0.018)	(0.017)	(0.016)
Observations	306	306	306
R-squared	0.682	0.791	0.791

SOURCE: 2008 through 2013 1-year ACS data.
NOTES: Shown are coefficients estimated from the specification of the determinants of state-year marriage rates. Each column presents results from a single regression. Unemployment rates come from the Bureau of Labor Statistics. Regressions are estimated using population weights. Specifications include state and year fixed effects, state-specific linear time trends, and state-year demographic controls. For the prior calendar year and prior 12 months estimates, the sample from which the marriage rate is estimated includes all unmarried women and women married in the prior calendar year or prior 12 months, as appropriate, ages 28–34 from the 2008 through 2013 1-year ACS data. For the constructed estimates, the sample from which the marriage rate is estimated includes all women ages 27–34 from the 2007 through 2013 1-year ACS data weighted by the proportion who were never married in the prior calendar year. Standard errors are clustered by state and shown in parentheses, *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Appendix

Appendix 1 Regression Results for Relationship between State-Year Unemployment and Log Marriage Rates

	Vital Statistics	ACS 2008-2013 Surveys	
	2007-2012 CY	Prior CY	Prior 12 Mo.
Unemployment rate	−0.010	−0.013	−0.013
	(0.008)	(0.010)	(0.011)
Observations	294	306	306
R-squared	0.970	0.936	0.943

SOURCE: 2007–2012 vital statistics data, 2008 through 2013 1-year ACS data.
NOTES: Shown are coefficients estimated from the specification of the determinants of state-year log marriage rates. Each column presents results from a single regression. Unemployment rates come from the Bureau of Labor Statistics. All regressions include state and year fixed effects, state-specific linear time trends, and state-year demographic controls and are weighted by the population estimates for unmarried females in each state-year cell. The sample for the vital statistics data does not include Nevada or Hawaii, Louisiana in 2006, and Oklahoma in 2000–2003. For vital statistics regressions, population estimates and demographic characteristics are from the CPS. The sample from which the ACS marriage rates are estimated includes all unmarried women and women married in the prior calendar year or prior 12 months ages 16 and older from the 2008 through 2013 1-year ACS data. Regressions are estimated using population weights. Standard errors are clustered by state and shown in parentheses, *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Appendix 2 Regression Results for Relationship between State-Year Unemployment and Log Marriage Rates Using the ACS

	Ages 16–27		Ages 28–34		Ages 35–64	
	Prior CY	Prior 12 Mo.	Prior CY	Prior 12 Mo.	Prior CY	Prior 12 Mo.
Unemployment rate	0.003	0.003	−0.035***	−0.036**	−0.011	−0.009
	(0.020)	(0.015)	(0.012)	(0.018)	(0.017)	(0.016)
Observations	306	306	306	306	306	306
R-squared	0.935	0.949	0.698	0.681	0.748	0.794

SOURCE: 2008 through 2013 1-year ACS data.
NOTES: Shown are coefficients estimated from the specification of the determinants of state-year log marriage rates. Each column presents results from a single regression. Unemployment rates come from the Bureau of Labor Statistics. Regressions are estimated using population weights. All regressions include state and year fixed effects, state-specific linear time trends, and state-year demographic controls and are weighted by the population estimates for unmarried females in each state-year cell. The sample from which the ACS marriage rates are estimated includes all unmarried women and women married in the prior calendar year or prior 12 months, as indicated, ages 16–64 from the 2008 through 2013 1-year ACS data, except as noted. Standard errors are clustered by state and shown in parentheses, *** $p<0.01$, ** $p<0.05$, * $p<0.1$.